

2024

COMPUTER SCIENCE — MINOR

Paper : MN-1

(Computer Fundamental and Digital Logic)

Full Marks : 75

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer *any five* questions :

2×5

- (a) Convert $(101101.101)_2$ into its decimal equivalent.
- (b) State and prove De-Morgan's theorem.
- (c) Distinguish between combinational and sequential circuit.
- (d) What is the difference between latch and flip-flop?
- (e) Define Serial-in-Parallel-out (SIPO) and Parallel-in-Serial-out (PISO) register.
- (f) What is Race-around condition?
- (g) Add: $(726)_8 + (321)_8$.
- (h) What is the main difference between CMOS and TTL?

Answer *any three* questions.

2. (a) Explain weighted and non-weighted code with suitable examples.

(b) How Binary Coded Decimal (BCD) differ from Grey Code in terms of representation and application?

3+2

3. Simplify the Boolean function :

$F(A, B, C, D) = \sum(0, 1, 2, 5, 8, 9, 10, 14)$ using Karnaugh map and implement it using NAND gates.

5

4. Design a 4-bit adder/subtractor circuit using logic gates and explain its working with suitable example.

5

5. Explain the working of a clocked/gated J-K flip-flop with the help of a truth table.

5

6. Design and explain the working of 4-bit UP ripple counter.

5

Please Turn Over

(1882)

Answer *any five* questions.

7. (a) Prove that the NOR gate is a universal gate by implementing XOR and XNOR logic gates. 6+4
(b) Perform 2's complement subtraction for $(10101)_2 - (1100)_2$ and explain the steps.
8. (a) What is cache memory? Explain its importance and how it improves processing speed. 5+5
(b) Differentiate between machine language and assembly language, using suitable examples.
9. (a) Implement Boolean function $F(A, B, C) = \sum(1, 2, 5, 7)$ using 4 : 1 multiplexer. 5+5
(b) Design a 1 : 4 De-multiplexer using logic gates and explain its working.
10. (a) Explain the working of BCD to Seven-Segment decoder with a neat labelled circuit diagram. 5+5
(b) Design a 4-2 priority encoder using logic gates and explain its operation.
11. (a) Design an asynchronous MOD-7 UP counter. 7+3
(b) Show the truth table and timing diagram.
12. (a) What is magnitude comparator? 2+(6+2)
(b) Design and explain the 2-bit magnitude comparator using logic gates. Draw the truth table.
13. (a) Draw and explain the working of a Master-Slave J-K flip-flop with a truth table. 7+3
(b) How the race around condition can be overcome by Master-Slave topology?
14. (a) Design a synchronous Decade UP counter and explain its operation. 8+2
(b) Draw the truth table of synchronous Decade UP counter.
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